2. Can be organized a Connections across the layers in standard topologies & among the units within a layer?
a) in feed forward manner
b) in feedback manner
c) both feed forward & feedback
d) either feed forward & feedback

Answer: d
Explanation: Connections across the layers in standard topologies can be in feed forward manner or in feedback manner but not both.

6. What is classification?
a) deciding what features to use in a pattern recognition problem
b) deciding what class an input pattern belongs to
c) deciding what type of neural network to use
d) none of the mentioned

Answer: b
Explanation: Follows from basic definition of classification.
7. What is generalization?
a) the ability of a pattern recognition system to approximate the desired output values for pattern vectors which are not in the test set.
b) the ability of a pattern recognition system to approximate the desired output values for pattern vectors which are not in the training set. c) can be either way
d) none of the mentioned

Answer: b
Explanation: Follows from basic definition of generalization.
8. What are models in neural networks?
a) mathematical representation of our understanding
b) representation of biological neural networks
c) both way
d) none of the mentioned
Answer: c
Explanation: Model should be close to our biological neural systems, so that we can have high efficiency in machines too.
10. Changing inputs affects what kind of dynamics directly?
a) synaptic
b) neural
c) activation
d) both synaptic & neural

Answer: c
Explanation: Activation dynamics depends on input pattern, hence any change in input pattern will affect activation dynamics of neural networks.

2. Can data be stored through weights?
a) yes
b) no

Answer: a
Explanation: Data can be stored through weights as in case of binary patterns.

3. How is pattern storage task generally accomplished?
a) by a feedback network consisting of processing units with non linear output functions
b) by a feedback network consisting of processing units with linear output functions
c) by a feedforward network consisting of processing units with non linear output functions
d) by a feedforward network consisting of processing units with linear output functions

Answer: b
Explanation: Pattern storage task generally accomplished by a feedback network consisting of processing units with non linear output functions.

4. The trajectory of the state is determined by?
a) activation dynamics
b) synaptic dynamics
c) both activation and synaptic dynamics
d) none of the mentioned

Answer: a
Explanation: The trajectory of the state is determined by activation dynamics.

1. How are input layer units connected to second layer in competitive learning networks? a) feedforward manner
b) feedback manner
c) feedforward and feedback
d) feedforward or feedback

Answer: a
Explanation: The output of input layer is given to second layer with adaptive feedforward weights.

2. Which layer has feedback weights in competitive neural networks? a) input layer
b) second layer
c) both input and second layer
d) none of the mentioned

Answer: b
Explanation: Second layer has weights which gives feedback to the layer itself.

3. What is the nature of general feedback given in competitive neural networks? a) self excitatory
b) self inhibitory
c) self excitatory or self inhibitory
d) none of the mentioned

Answer: a
Explanation: The output of each unit in second layer is fed back to itself in self – excitatory manner

4. What consist of competitive learning neural networks?
a) feedforward paths
b) feedback paths
c) either feedforward or feedback
d) combination of feedforward and feedback

Answer: d
Explanation: Competitive learning neural networks is a combination of feedforward and feedback connection layers resulting in some kind of competition.

5. What conditions are must for competitive network to perform pattern clustering? a) non linear output layers
b) connection to neighbors is excitatory and to the farther units inhibitory
c) on centre off surround connections
d) none of the mentioned fulfils the whole criteria

Answer: d
Explanation: If the output functions of units in feedback layer are made non-linear , with fixed weight on-centre off-surround connections, the pattern clustering can be performed.

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Explanation: If the output functions of units in feedback layer are made non-linear , with fixed weight on-centre off-surround connections, the pattern clustering can be performed.

6. What conditions are must for competitive network to perform feature mapping? a) non linear output layers
b) connection to neighbors is excitatory and to the farther units inhibitory
c) on centre off surround connections
d) none of the mentioned fulfils the whole criteria

Answer: d
Explanation: If condition in a, b, c are met then feature mapping can be performed.

7. If a competitive network can perform feature mapping then what is that network can be called?
a) self excitatory
b) self inhibitory
c) self organization
d) none of the mentioned

Answer: c
Explanation: Competitive network that can perform feature mapping can be called as self organization network.

8. What is an instar?
a) receives inputs from all others
b) gives output to all others
c) may receive or give input or output to others
d) none of the mentioned

Answer: a
Explanation: An instar receives inputs from all other input units.

9. How is weight vector adjusted in basic competitive learning?
a) such that it moves towards the input vector
b) such that it moves away from input vector
c) such that it moves towards the output vector
d) such that it moves away from output vector

10. The update in weight vector in basic competitive learning can be represented by? a) w(t + 1) = w(t) + del.w(t)
b) w(t + 1) = w(t)
c) w(t + 1) = w(t) – del.w(t)
d) none of the mentioned

Answer: a
Explanation: The update in weight vector in basic competitive learning can be represented by w(t + 1) = w(t) + del.w(t).

6. What does GRNN do?
a) function approximation task
b) pattern classification task
c) function approximation and pattern classification task
d) none of the mentioned

Answer: a

1. For what purpose energy minima are used?
a) pattern classification
b) patten mapping
c) pattern storage
d) none of the mentioned

Answer: c
Explanation: Energy minima are used for pattern storage.
2. Is it possible to determine exact number of basins of attraction in energy landscape? a) yes
b) no

Answer: b
Explanation: It is not possible to determine exact number of basins of attraction in energy landscape.
3. What is capacity of a network?
a) number of inputs it can take
b) number of output it can deliver
c) number of patterns that can be stored
d) none of the mentioned

Answer: c
Explanation: The capacity of a network is the number of patterns that can be stored.

4. Can probability of error in recall be reduced?
a) yes
b) no

Answer: a
Explanation: Probability of error in recall be reduced by adjusting weights in such a way that it is matched to probability distribution of desired patterns.
5. Number of desired patterns is what of basins of attraction?

a) dependent
b) independent
c) dependent or independent
d) none of the mentioned

Answer: b
Explanation: Number of desired patterns is independent of basins of attraction.

6. What happens when number of patterns is more than number of basins of attraction? a) false wells
b) storage problem becomes hard problem
c) no storage and recall can take place
d) none of the mentioned

Answer: b
Explanation: When number of patterns is more than number of basins of attraction then storage problem becomes hard problem.

7. What happens when number of patterns is less than number of basins of attraction? a) false wells
b) storage problem becomes hard problem
c) no storage and recall can take place
d) none of the mentioned

Answer: a
Explanation: False wells are created when number of patterns is less than number of basins of attraction.

8. What is hopfield model?
a) fully connected feedback network
b) fully connected feedback network with symmetric weights
c) fully connected feedforward network
d) fully connected feedback network with symmetric weights

Answer: b
Explanation: Hopfield model is fully connected feedback network with symmetric weights.

9. When are false wells created?
a) when number of patterns is more than number of basins of attraction
b) when number of patterns is less than number of basins of attraction
c) when number of patterns is same as number of basins of attraction
d) none of the mentioned

Answer: b

Explanation: False wells are created when number of patterns is less than number of basins of attraction.

10. When does storage problem becomes hard problem?

a) when number of patterns is more than number of basins of attraction
b) when number of patterns is less than number of basins of attraction
c) when number of patterns is same as number of basins of attraction
d) none of the mentioned

Answer: a
Explanation: When number of patterns is more than number of basins of attraction then storage problem becomes hard problem

1. Logistic regression in vanilla form can be used to solve multiclass classification problems. a) True
b) False

Answer: b
Explanation: Logistic regression is said to be in vanilla form when not customized from their original form, meaning that they are used without any customizations or updates applied to them. In its vanilla form logistic regression is used to do binary classification and not for multiclass classification.

2. Multiclass classification with logistic regression can be done through the one-vs-rest scheme.

a) True
b) False

Answer: a
Explanation: Multiclass classification with logistic regression can be done either through the one-vs-rest scheme in which for each class a binary classification problem of data belonging or not to that class is done, or changing the loss function to cross- entropy loss.

3. The goal of multiclass classification is to construct a function which, given a new data point, will correctly predict the class to which the new point belongs. a) True
b) False

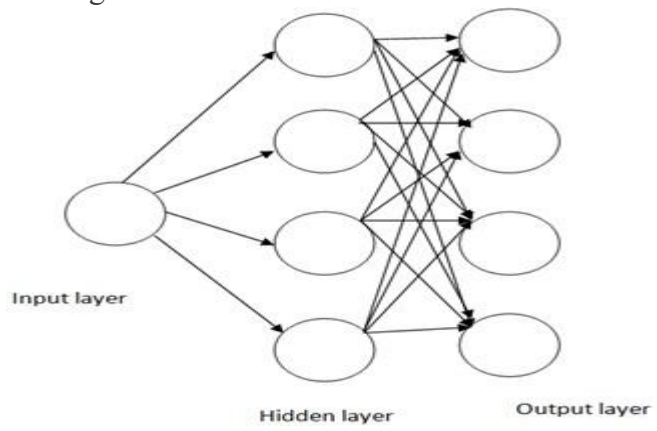
Answer: a
Explanation: In multiclass classification each training point belongs to one of N different classes. The goal is to construct a function which, given a new data point, will correctly predict the class to which the new point belongs. The points can belong to multiple categories, but a point can belong to only in one category.

4. Softmax layer is one of the best options as the output layer in a multiclass classification implementation. a) True
b) False

Answer: a
Explanation: While implementing a multiclass classification problem the important part is the choice of the output layer. The usual choice for multi-class classification is the softmax layer.

The softmax function is a generalization of the logistic function that “squashes” a KKdimensional vector z of arbitrary real values to a **K** – dimensional vector **σ (z)** of real values in the range [0, 1], that add up to 1.

5. The given neural network cannot be used to solve multiclass classification problems.



a) True
b) False

Answer: b
Explanation: The given figure shows a single input – multi output neural network with two hidden layers. And here it gets input from single source and produce the multiple outputs as in a multiclass classification problem to find out the output class of the given data point. So clearly this neural network can be used for multiclass classification problems.

6. One vs. all becomes increasingly inefficient as the number of classes rises. a)
True
b) False

Answer: a
Explanation: One vs. all involves the training of N-binary classifiers where each of them is designed to recognize a particular class. This approach is fairly reasonable when the total number of classes is small, but becomes increasingly inefficient as the number of classes rises. We can create a significantly more efficient one-vs.-all model with a deep neural network in which each output node represents a different class

1. Can Invariances be built as static functions in the structure? a)
yes
b) no

Answer: b
Explanation: Invariances have to be dynamically estimated from data.

2. What is the objective of associative memories?
a) to store patters
b) to recall patterns
c) to store association between patterns

d) none of the mentioned

Answer: d
Explanation: The objective of associative memories is to store association between patterns for later recall of one of patterns given the other.

3. Is it possible to capture implicit reasoning process by pattern classification network? a) yes
b) maybe
c) no
d) cannot be determined

Answer: a
Explanation: For example neural network for contract bridge game.

4. Are classification methods based on correlation matching using moment features useful for problems of handwritten characters? a) yes
b) no

Answer: b
Explanation: Because different parts of handwritten characters are deformed differently.

5. Associative memory, if used in feedback structure of hopfield type can function as? a) data memory
b) cluster
c) content addressable memory
d) none of the mentioned

Answer: c
Explanation: Associative memory, if used in feedback structure of hopfield type can function as content addressable memory.

6. In feedforward network, the associations corresponding to input – output patterns are stored in?
a) activation state
b) output layer
c) hidden layer
d) none of the mentioned

Answer: d
Explanation: In feedforward network, the associations corresponding to input – output patterns are stored in weights of the network.

7. Which is one of the application of associative memories?
a) direct pattern recall
b) voice signal recall

c) mapping of the signal
d) image pattern recall from noisy clues

Answer: d
Explanation: The objective of associative memories is to store association between patterns for later recall of one of patterns given the other, so noisy versions of the same image can be recalled.

8. How can optimization be applied in images?
a) by use of simulated annealing
b) by attaching a feedback network
c) by adding an additional hidden layer
d) none of the mentioned

Answer: a
Explanation: Optimization be applied in images by use of simulated annealing to formulate the problem as energy minimization problem.

9. In control applications, how many ways are there to control a plant? a)
1
b) 2
c) 4
d) infinite

Answer: b
Explanation: Open loop and feedback loop are the two ways.

10. Neuro – Fuzzy systems can lead to more powerful neural network? a)
yes
b) no
c) may be
d) cannot be determined

Answer: a
Explanation: If fuzzy logic is incorporated into conventional ANN models, more powerful systems can be created